Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/08/2022
PI Name:	Stitzel, Joel Ph.D.		
Project Title:	Investigation of Occupant Injury Risk in the Soyuz Vehicle and Comparison to Commercial Crew Designs		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jstitzel@wakehealth.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	336-716-5597
Organization Name:	Wake Forest University		
PI Address 1:	Biomedical Engineering		
PI Address 2:	1 Medical Center Boulevard		
PI Web Page:			
City:	Winston-Salem	**State:**	NC
Zip Code:	27157-0001	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	10/01/2020	**End Date:**	01/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/31/2022, per NSSC information (Original end date was 9/30/2021)--Ed. 8/26/21		
Key Personnel Changes/Previous PI:	Editor's Note: As per the Principal Investigator (PI), Jeffrey Somers, Ph.D. was added to the project during this reporting period. Dr. Somers is with KBR/NASA Johnson Space Center. (Ed., 6/9/22).		
COI Name (Institution):	Abercromby, Andrew Ph.D. (NASA Johnson Space Center) Gayzik, Francis Ph.D. (Wake Forest University) Jones, Derek Ph.D. (Elemance, LLC) Newby, Nathaniel M.S. (KBR/NASA Johnson Space Center) Weaver, Ashley Ph.D. (Wake Forest University) Somers, Jeff (NASA)		
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	Currently, United States (US) astronauts return to Earth aboard the Soyuz spacecraft. Recent evidence has shown that the occurrence of injuries in crewmembers landing in the Soyuz spacecraft is higher than predicted by current models. There are several possible explanations for this trend: the landing impacts are harder than predicted, current models do not accurately capture the true injury risk, spaceflight deconditioning decreases the crewmembers' tolerance to impact, or some combination of the three. In addition, NASA and commercial companies are developing three new capsule-type spacecraft in which crewmembers will be subjected to a landing impact at the end of the mission. We need an accurate understanding of the mechanisms of landing impact injury to predict risk of crewmember injury in these vehicles. To improve our understanding of Soyuz crewmember injury risk, as well as future crewmembers onboard the three new designs (Orion, SpaceX Dragon2, and Boeing CST-100), we propose a computational study whose key objectives are to assess the risk of injury to Soyuz crewmembers and compare the Soyuz occupant protection design to the three future vehicle designs. Using mid-size male finite element (FE) models of a human with varying levels of complexity, Soyuz landings will be simulated and responses from the model will be analyzed to assess injury risk. These injury risk predictions will be correlated with actual injury outcomes in crewmembers captured in the Human Research Program (HRP)-funded Soyuz Landing Injury Risk Characterization Study (Principal Investigator: Newby). Soyuz human FE injury risk predictions will also be compared to our previous human FE simulations of the three modern vehicle designs from the HRP-funded ATD (Anthropomorphic Test Dummy) Injury Metric Sensitivity and Extensibility Study (Principal Investigator: Somers; Co-Investigators: Stitzel, Gayzik, Weaver, Newby). This will greatly aid NASA to estimate crew injury risk as a design variable for future missions.
Task Description:	By assessing Soyuz landing impacts with a human FE model in flight-like conditions and comparing the Soyuz occupant protection design to other modern vehicle designs, this proposal directly addresses the Human Research Roadmap (HRR) Risk of Injury Due to Dynamic Loads and would provide additional evidence to characterize the risk as well as contribute to addressing three Occupant Protection gaps (OP-101: We do not understand the risk of injury associated with crewed vehicle landings and how this risk relates to the desired acceptable risk; OP-301: We do not have an identified, validated and standardized approach for vehicle instrumentation and biodynamic data collection, and predictive analytic biodynamic modeling that would allow for specific risk injury prediction by mission-phase, crew functionality post-landing, and vehicle design; and OP-401: We do not know the extent to which multiple spaceflight hazards (e.g., spaceflight deconditioning, bone loss, radiation exposure, altered gravity) may interact to synergistically decrease injury tolerance for off-nominal dynamic landing loads, increasing risk to crew's performance in mission-completing actions immediately after landing.). The project will yield novel scientific knowledge on crewmember injury risk associated with landing events in the Soyuz, Orion, Dragon2, and CTS-100 vehicles. The work will also produce validated analytical tools including a Soyuz seat FE model with mid-size male human body FE models which can be used to verify occupant safety of crewmembers in impact conditions. By achieving the aims of the proposed work, the study will advance our overall understanding of crewmember injury risk due to dynamic loads encountered in spacecraft landings, and produce validated computational models which can be used to predict the associated occupant injury risk when the Soyuz and modern spacecraft vehicle designs are subjected to various dynamic loading events.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Landings in the Soyuz Vehicle have generated more injuries than models have predicted, and the reason for this remains unknown. Presently, three new capsule-type spacecraft – the Orion, SpaceX Dragon2, and the Boeing CST-100 – are in various stages of development. To better understand the injury risk to Soyuz crewmembers, and to mitigate risk of future injuries on board the new vehicles, a finite element (FE) modeling study was conducted. At the outset of this study, previously developed FE models were available for the seats in the three new vehicles, whereas the Soyuz environment has not been modeled in an extensible manner. Thus, the development of a model of the Soyuz seat was necessary to facilitate the comparisons of all seat types. Ultimately, the goal of this project was to generate data that will help answer the question: "Why was the rate of injury in the Soyuz vehicle higher than expected?". The study consisted of two specific aims: (1) Identify expected injury risks associated with nominal and off-nominal landing scenarios in the Soyuz, and quantify differences in injury risk predicted by human body models (HBMs) of varying complexity. (2) Examine differences in injury risk to crews of Soyuz and modern vehicles as it relates to seat and restraint design and loading combinations. In conclusion, this study developed a finite element model of the Soyuz seat that is usable for future studies. The detailed and simplified Global Human Body Models Consortium (GHBMC) human body models were compared in a limited set of Soyuz landing cases. Little difference was observed for the two models across the full body, with exceptions for head rotational velocity, chest compression, and lumbar moments. Therefore, unless a future study is focused specifically in those regions, the benefits of the significantly faster run-time of the simplified GHBMC can be taken advantage of. It was demonstrated that the lack of a side head guard in the Soyuz produced higher neck tension values, which may explain the increased presence of whiplash-like injuries in the real-world landings. However, the elevated lumbar spine injury rate was unexplained. Therefore, the authors believe that many of the elevated injury occurrences in the Soyuz landings may be attributable to higher impact kinematics than expected. Future work should be focused on reconstructing these landings, and take advantage of the newly developed FE model of the Soyuz seat. As the modern vehicles continue to be used in American spaceflight, instrumentation should be included in the capsules so that landings are documented in real time. If an injury occurs, the impact should be simulated with an appropriately scaled HBM and seat to understand the mechanism of injury. If an obvious design change is clear, and doesn't affect operation, that change can potentially be made to protect future occupants.
Bibliography Type:	Description: (Last Updated:)